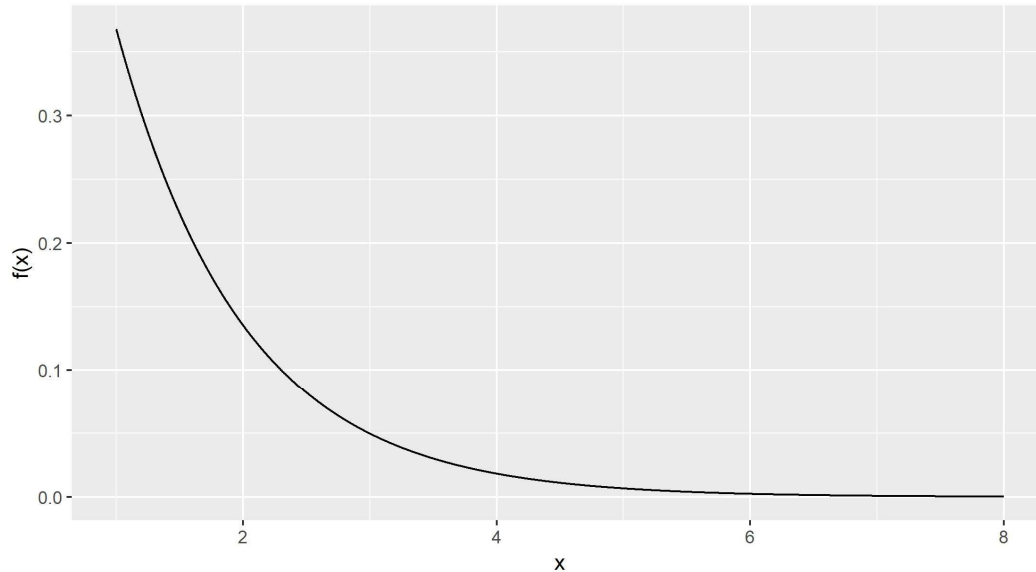


Exponential distribution

Exponential distribution



$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$E(X) = \frac{1}{\lambda}$$

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

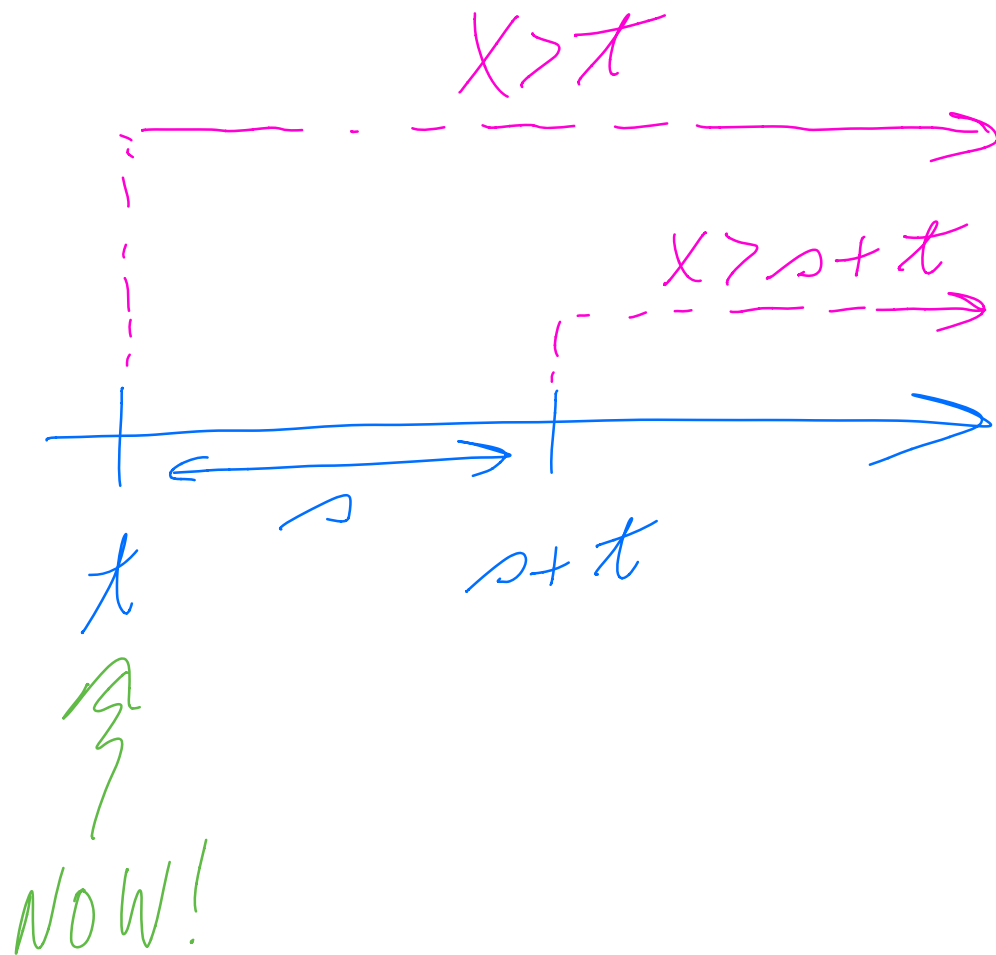
Exponential distribution – properties

- The exponential distribution is **memoryless**, i.e.,

$$P(X > s + t | X > t) = P(X > s) \text{ for all } s, t \geq 0$$

- If the duration of a telephone call is exponentially distributed then, at each time instant, the time to complete the call is independent of the time already spent in the call – Is this realistic?
- Example where $P(X > s + t | X > t)$ goes down when t goes up: lifetime of a car
- Example where $P(X > s + t | X > t)$ goes up as when t goes up: UNIX job CPU lifetimes

$$P(X > s+t \mid X > t) = P(X > s)$$



$$P(X > s+t | X > t) = \frac{P(X > s+t, X > t)}{P(X > t)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(X > s+t)}{P(X > t)}$$

Since $P(X > s+t | X > t) = P(X > s)$
it results

$$P(X > s+t) = P(X > t) P(X > s)$$

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

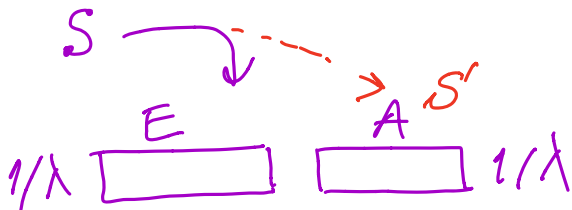
$$P(X > x) = e^{-\lambda x}, \quad x \geq 0$$

$$e^{-\lambda(s+t)} = e^{-\lambda s} \cdot e^{-\lambda t} = e^{-\lambda(s+t)}$$

yes!

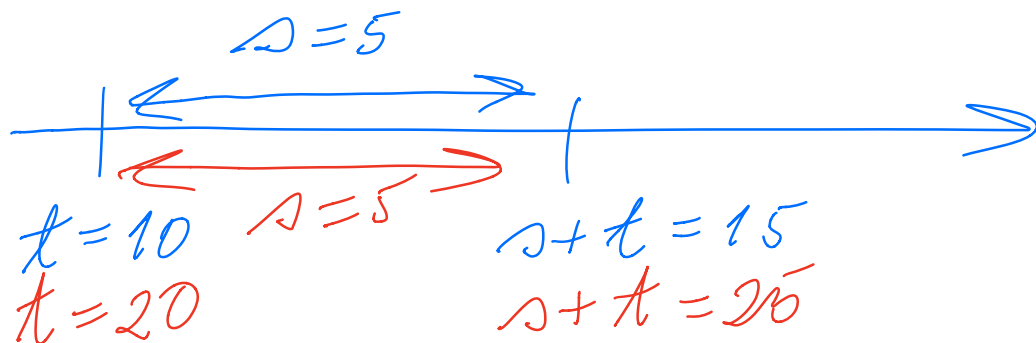
Exponential distribution – properties

- The duration of a telephone call is exponentially distributed with average 10 minutes. The probability that the call lasts more than 15 minutes given that it already lasted 10 minutes is $P(X > 5) \approx 0.604$; the probability that the call lasts more than 25 minutes given that it already lasted 20 minutes is the same!
- Consider a bank with two employees. When Mr. Saravanan arrives at the bank finds that Mr. Evariste is being served by one employer and Mr. Abubakar by the other. Suppose that Mr. Saravanan will start being served as soon as Mr. Evariste or Mr. Abubakar leave the bank. The service time of each employer is exponentially distributed with average $1/\lambda$. What is the probability that Mr. Saravanan is the last one to leave the bank?



$$P_S = 0.5$$

$$P(X > 15 | X > 10) = P(X > 5)$$



$$\begin{aligned}
 P(X > 5) &= 1 - P(X \leq 5) = 1 - (1 - e^{-\lambda 5}) \\
 &= e^{-\lambda 5} = e^{-0.5} = 0.604
 \end{aligned}$$

$$1/\lambda = 10$$

$$\begin{aligned}
 P(X > 25 | X > 20) &= P(X > 5) \\
 &= 0.604
 \end{aligned}$$

Exponential distribution – properties

- If X_1 and X_2 are independent and exponentially distributed r.v. with means $1/\lambda_1$ e $1/\lambda_2$ respectively, then

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

- A stereo system has two main parts: radio and speakers. If the lifetime of the radio is exponentially distributed with mean 1000 hours and the lifetime of the speakers is exponentially distributed with mean 500 hours, and both lifetimes are independent, what is the probability that a failure in the system is caused by a failure in the radio?

$X_1 \equiv$ lifetime of radio
 $X_2 \equiv$ " of speakers

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$\lambda_1 = 1/1000 \quad \lambda_2 = 1/500$$

$$P(X_1 < X_2) = \frac{1/1000}{1/1000 + 1/500} = \frac{1}{3}$$